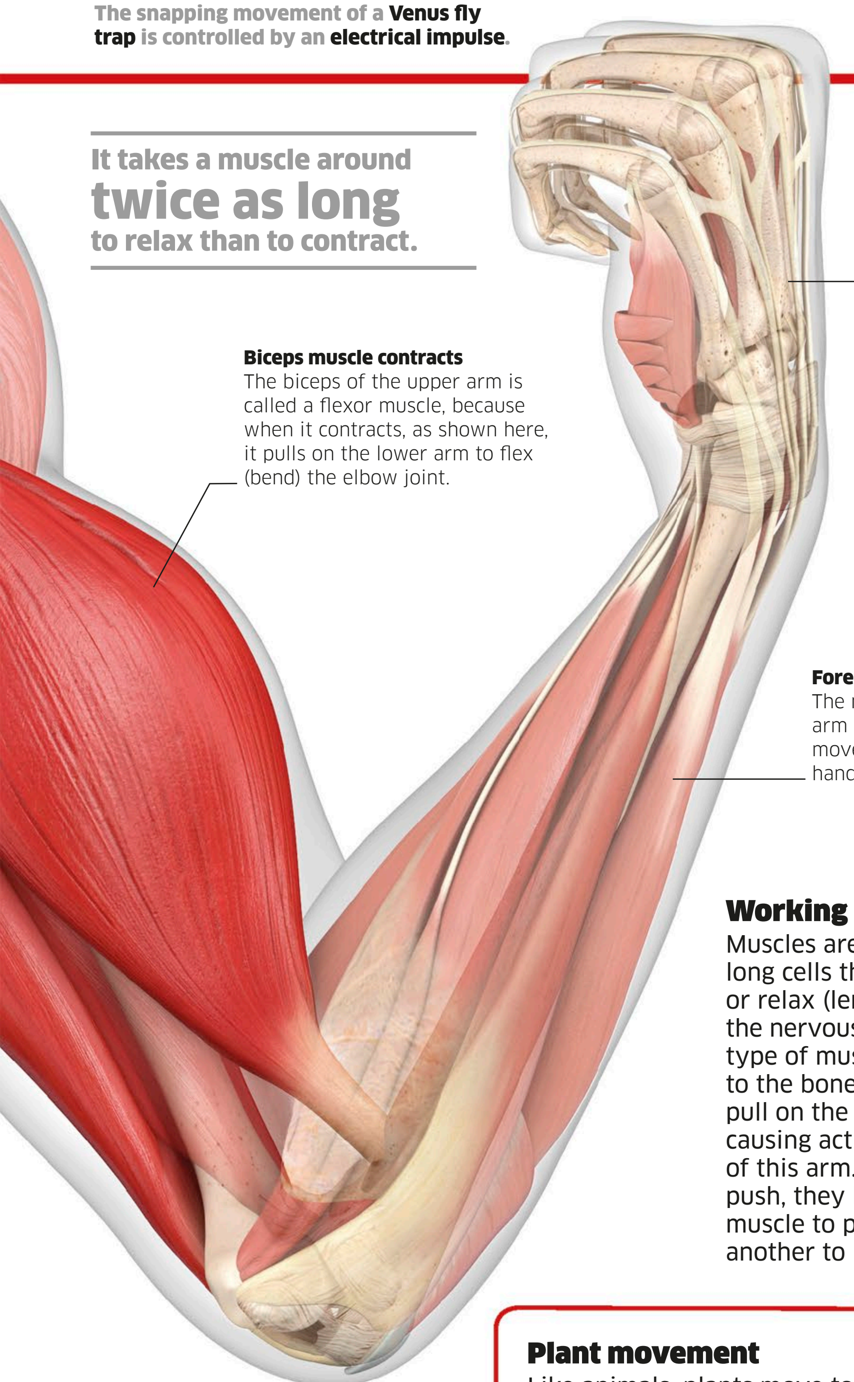
The snapping movement of a **Venus fly trap** is controlled by an **electrical impulse**.

Heart muscle is **the only muscle that can spontaneously contract** without being triggered by a nerve impulse.

It takes a muscle around **twice as long** to relax than to contract.

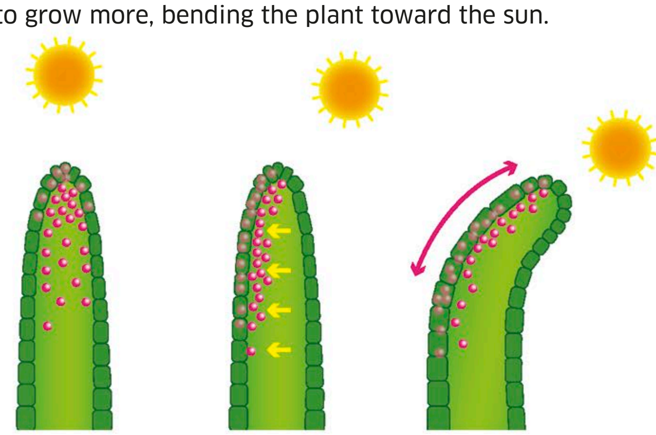


Working in pairs

Muscles are made up of bundles of long cells that either contract (shorten) or relax (lengthen) when triggered by the nervous system. The most common type of muscles are those connected to the bones of the skeleton. They pull on the bones when they contract, causing actions like the movement of this arm. Because muscles cannot push, they have to work in pairs—one muscle to pull the arm upward and another to pull it back down.

Plant movement

Like animals, plants move to make the most of their environment. The shoot tips of plants are especially sensitive to light and can slowly bend toward a light source. A chemical called auxin (which regulates growth) encourages the shadier side of the shoot to grow more, bending the plant toward the sun.

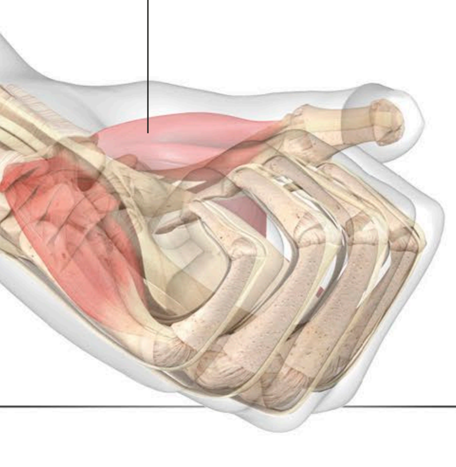


Auxin (pink) produced in the shoot tip spreads down through the shoot, making it grow upward.

When light shines from one direction, auxin moves to the shadier side.

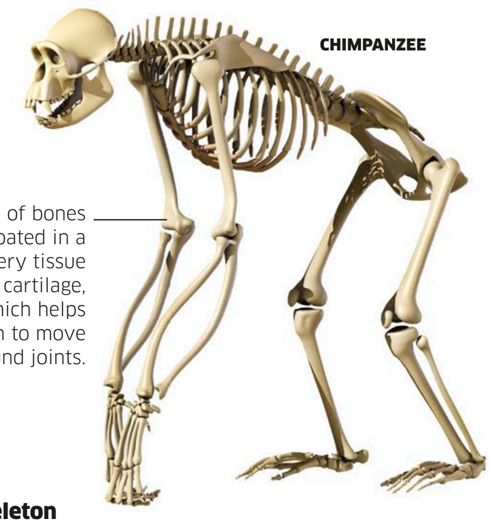
On the shadier side, the auxin stimulates the plant cells to grow bigger, so that the shoot bends toward the light.

Rotating the arm
As well as pulling the lower arm toward it, the biceps can also rotate the forearm so the palm of the hand faces upward.

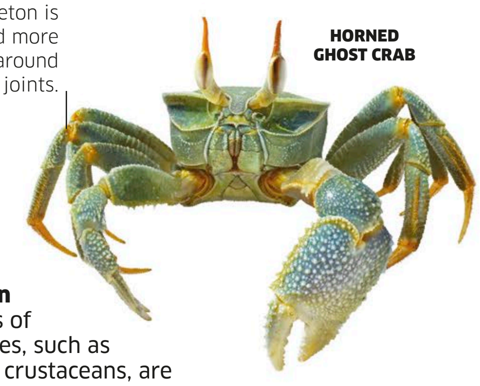


Support structures

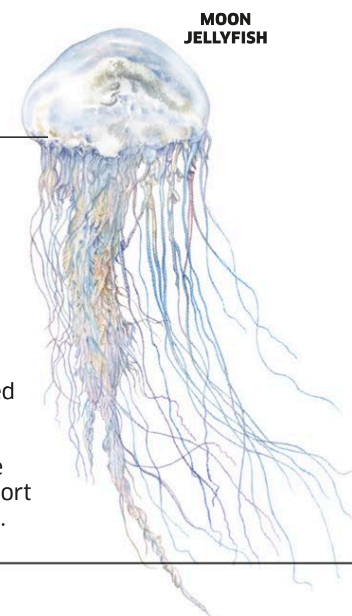
Animals have a skeleton to support their bodies and protect their soft organs. This is especially important for large land-living animals that are not supported by water. Skeletons also provide a firm support for contracting muscles, helping animals to have the strength to move around.



Endoskeleton
Vertebrate animals—including fish, amphibians, reptiles, birds, and mammals—have a hard internal skeleton within their bodies. The muscles surround the skeleton and pull on its bones.



Exoskeleton
Many types of invertebrates, such as insects and crustaceans, are supported by an external skeleton that covers their body like a suit of armor with muscles inside. Exoskeletons cannot grow with the rest of the body, so must be periodically shed and replaced.



Hydroskeleton
Some kinds of soft-bodied animals, such as sea anemones and earthworms, are supported by internal pouches that stay firm because they are filled with fluid. These water-filled pouches support the muscles as they move.